

How Heterogeneous Systems Evolved and the Challenges, Going Forward

In laymen's terms, computer architecture refers to how a computer is designed and how software interacts with it. With the demands on computing power increasing, it is a natural corollary that computer architecture should also evolve. This article highlights the challenges and opportunities in designing heterogeneous systems.

The emergence of mobility, social networking, campus networking and edge computing has brought with it an essential need to process varieties of data – within and outside of enterprise data centres. Traditional computing needs to cope not only with large volumes of data, but also with the wide variety and velocity of data. The need to handle data velocity, variety and volume exists across the entire computing landscape, starting from handheld devices to supercomputing infrastructure.

Purpose-built computers that are armed with FPGAs (field-programmable gate arrays) or accelerators and off-loading, assist in meeting some of the special data processing needs of today, such as scene detection from a remote camera feed, image search, and speech recognition. Faced with humongous Big Data processing needs—be it text, images, audio, video or data from billions of sensors—traditional computing architectures find it difficult to handle the volume, velocity and variety of the data. While it is true that over-provisioned general-purpose computing infrastructure or expensive specialised infrastructure can get the job done somehow, traditional computing infrastructure has shown itself to be inefficient in meeting the needs of Big Data. For efficient Big Data computing at the right cost, a paradigm shift is needed in computer architecture.

In today's computing landscape, there are certain basic technology drivers that have reached their limits. Shrinking transistors have powered 50 years of advances in computing, as Moore's Law, named after Intel co-founder Gordon Moore says "number of transistors in a dense integrated circuit doubles approximately every two years." But for some time now, making transistors smaller has not made them more

energy-efficient. So while the benefits of making things smaller have been decreasing, the costs have been rising. This is in large part because the components are approaching a fundamental limit of smallness—the atom.

On the other hand, disruptive technology trends like non-volatile memory and photonic interconnects call for revamping the existing, traditional architectures. There has also been a move by certain parts of the industry to adopt a

data-centric computing paradigm, as opposed to the traditional CPU or compute-centric approach, though this nascent shift has largely remained at the level of academic research.

At the same time, there is a constant demand for improved performance to enable compelling new user experiences. New use cases like molecular modelling, photo-realistic ray tracing and modelling, seismic analysis,

augmented reality and life sciences computing have really questioned how effective these traditional compute units are, in today's context. To address the twilight of Moore's Law and to navigate this complex set of requirements, the computer industry needs an approach that promises to deliver improvements across four vectors—power, performance, programmability and portability. This has led to the industry exploring the area of heterogeneous systems in order to meet the demands of newer compute- and data-centric applications.

In this article, we discuss the evolution of heterogeneous systems, as well as the challenges and opportunities in this field, with regard to compute units, interconnects, software architecture, etc. Given the vastness and depth of this topic, we intend to provide a brief overview of heterogeneous systems in this issue, followed by a detailed in-depth

